

RFI response regarding the national

AI Action Plan

by

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1. The dangers of using AI

While AI has demonstrated a potential for significant beneficial impact in many market and government sectors, it has also been demonstrated to suffer from bias and security vulnerabilities. These are serious concerns for high-risk applications. Defending and robustifying AIs is not a trivial matter, particularly if one does not want to impede innovative applications.

The vulnerabilities of an AI are likely exacerbated by excess parameterization, i.e., models which are too large. For decades, machine learning researchers have understood the risks of over-parameterization, which may cause excessive error (model variance) due to overfitting. Indeed, modern deep learning frameworks have techniques (such as random dropout) which are intended to address this problem.

However, some “research” has questioned whether overfitting is a problem for applications requiring very large models (like LLMs); for example, the so-called “double-descent hypothesis”. This research has been used to push for ever-larger models, both driving and justifying the trend toward huge data-centers (clouds), consisting of many thousands of enormous and expensive hardware accelerators (e.g., Nvidia H100 GPUs).

So, the release of Deepseek earlier this year by the Chinese was a surprise to some. Deepseek is an open-source general-purpose chatbot which was trained on publicly available data using lower-end, inexpensive GPUs. Little about Deepseek’s design is truly novel. Compared to the latest closed-source ChatGPT developed by OpenAI in the USA, Deepseek is a smaller model with arguably better performance and which was produced at a fraction of the cost.

2. The dangers of overfunding, particularly AI

Just as excess model parameterization, excess research funding can yield negative results, particularly for AI related research, which is a highly data-dependent area involving a lot of ad hoc decision-making through trial-and-error experimentation.

The four main “AI conferences” (NeurIPS, ICML, AAAI, ICLR) annually receive a total of about 50,000 submissions, of which a total of about 10,000 are accepted (about 10% of accepted papers are orally presented; the rest are “presented” only in large poster sessions). At this scale, these are not genuine academic conferences but more like money-making annual conventions which mass-produce “accepted research

articles". Also, they can be more cynically characterized as cesspits of money (much of it from the federal government), influence peddling, and poor-quality research.

This is typical of other heavily funded, applied-research communities. A representative of the Association of Computing Machinery (ACM) presented a slide at ACM CCS 2024 (one of the three "top" conferences in cyber security) chronicling corruption in its putatively double-blind peer-review process (including a case where an author was caught uploading a review for his own paper). Note that such papers appear as peer-reviewed work-product in annual reports (to federal agencies) of research grants. Leadership in federal research-funding agencies cannot deny they are aware of these ethical problems, which may be construed as fraud. For that matter, why should systematic peer-review corruption be limited to research conferences and not be present in the proposal-review panels where the stakes are much higher?

3. Recommendations regarding AI Research Funding

We suggest targeted investments in secure and robust AI, which is a cross-disciplinary area between AI/ML and cyber security. Smaller grants will be most impactful, while the numerous and very large-scale AI centers have proven a waste of taxpayers' money.

A lower limit to the annual number of papers per researcher needs to be set to promote research quality (capping conference registration costs and publication charges allowed on federal grants can also help with this). Just as one example, a recent high-profile conference warned the authors of submitted papers that any submissions **beyond 25 by the same author (!)** will be automatically rejected. But allowing 25 papers submitted by the same author to a single conference is obviously about bean counting, not quality.

The balance of research investments in AI can be redirected to graduate-student scholarships. This could be a far more efficient use of precious research funds. Universities need to be encouraged to refocus on properly educating our student scientists, engineers and computer scientists, rather than running degree mills and paper mills. Despite much higher cost of education, our average undergraduates are at present significantly inferior compared to foreign undergraduates (who often have a far better grasp of the basics). AI is exacerbating these negative trends.

Though there are some accounts of these problems in the public media (mass retractions of published research articles papers, co-citation cartels, fake-paper authorship marketplaces, poor literacy and numeracy rates of average college graduates, etc.), there has been in the past very little leadership from government on the issue of research ethics and generally holding very well compensated university administrators more accountable. This needs to be a focus of the new federal leadership in the context of an AI Action Plan.

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